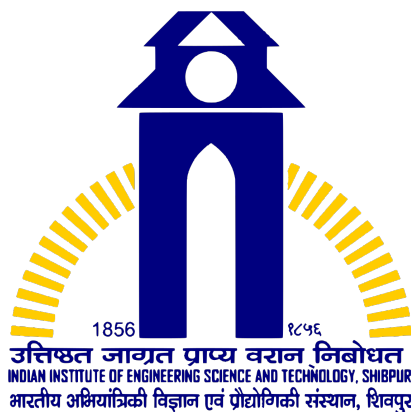




SIX LANE ELEVATED KONA EXPRESSWAY FROM KM 0.145 TO KM 7.337 OF NH-117 IN THE STATE OF WEST BENGAL

A WINTER INTERNSHIP REPORT-(2025-2026)

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Abstract

This report presents the learning outcomes and practical exposure gained during a one-month winter internship conducted under the **National Highways Authority of India (NHAI)**, with project execution managed by **GPT Group** and construction work contracted to **Rail Vikas Nigam Limited (RVNL)**. The internship provided extensive insight into the planning, execution, testing, and quality control of large-scale highway infrastructure projects, integrating theoretical knowledge with on-site practices in both structural and geotechnical domains.

During the training period, various on-site construction activities were observed, including concrete pouring for prestressed members, pre-tensioning operations, and the casting and erection of prestressed concrete (PSC) girders. The study emphasized the behavior of prestressed elements, stress distribution, load transfer mechanisms, and safety factors adopted in design and construction. Foundation works, such as pile casting, pile driving, and piling arrangements executed by GPT Group under RVNL supervision, were analyzed to understand foundation stability and load-bearing behavior.

Field and quality control tests conducted on-site included the **Standard Penetration Test (SPT)** for soil investigation, **slump and compaction factor tests** for fresh concrete workability, and **cube compressive strength tests** for concrete quality assessment. Additional observations encompassed reinforcement inspections, concrete cover verification, alignment and level checks, and monitoring of piling operations. Site safety inspections and audits were also performed to ensure compliance with prescribed safety standards during construction activities.

The experience provided hands-on exposure to real-time construction processes, testing procedures, and safety management, effectively bridging the gap between academic learning and professional practice. The internship enhanced technical understanding, field awareness, and professional development in highway and structural engineering projects executed by GPT Group under the supervision of RVNL.

Keywords: Highway construction, prestressed concrete, foundation works, quality control, field testing, site safety.

Contents

1	Introduction	4
2	Objectives of the Internship	4
3	Description about the Organization	4
4	Project Description	4
4.1	Project Location and Scope	4
4.2	Alignment, Length, and Structural Components	4
4.3	Construction Methodology	5
5	Casting Yard Operations	5
5.1	Layout and Function of Casting Yard	5
5.2	Materials Used (Cement, Aggregates, Steel, Admixtures)	5
5.3	Pre-tensioning and Post-tensioning System	5
5.4	PSC Girder Casting Process	6
5.5	Curing, De-shuttering, and Storage of Girders	6
6	Pre-stressed Concrete (PSC) Girders	7
6.1	Cable Layout and Anchorage System	8
6.2	Stress-Strain Behavior in Upward and Downward Loading Cases	8
6.3	Quality Control Measures	9
6.4	Stressing and De-stressing Operations and Equipment	9
6.5	Stressing Calculations and Procedure	10
6.5.1	Design Stress and Jacking Force	10
6.5.2	Elongation Calculation	10
6.5.3	Site-Measured Elongation and Verification	11
6.5.4	Permissible Tolerances	11
6.5.5	Stressing Sequence and Safety Measures	11
6.6	Haunching and Reinforcement Detailing	12
6.7	Girder Erection Operations	12
6.7.1	Risk Control Matrix for Girder Erection (HIRA)	13
6.7.2	Girder Erection Procedure Flowchart	14
7	Pile Foundation Works	15
7.1	Types of Piles Observed	15
7.2	Piling Equipment and Methods	15
7.3	Boring, Concreting, and Reinforcement	15
7.4	Pile Load Tests and Integrity Tests	15
7.5	Standard Penetration Test (SPT): Procedure, Requirements, and Interpretation	16
7.5.1	Purpose of SPT	16
7.5.2	Equipment Used	16
7.5.3	SPT Procedure	16
7.5.4	Corrections Applied to N-Value	17
7.5.5	Interpretation of SPT N-Value	18
7.5.6	Field Observations from Site Notes	18
7.5.7	Importance of SPT in Pile Design	18
8	Reinforcement Detailing of Pile	18
8.1	General Description	18
8.2	Pile Geometry and Size	19
8.3	Longitudinal Reinforcement	19

8.4	Lateral Reinforcement (Ties / Helical Reinforcement)	19
8.5	Pile Head Reinforcement and Connection with Pile Cap	19
8.6	Corrosion Protection and Durability Measures	19
8.7	Reinforcement Summary Table	20
8.8	Construction and Placement Notes	20
9	Superstructure Construction	20
9.1	Pier and Pier Cap Construction	20
9.2	Bearing Installation	21
9.3	Deck Slab and Wearing Course	21
9.4	Expansion Joints and Drainage Provisions	21
10	Laboratory Testing	21
10.1	Concrete Cube Compressive Strength Test	21
10.2	High-Frequency Motor Concrete Cube Vibrator	22
10.3	Aggregate Impact Value (AIV) Test	22
10.4	Concrete Permeability Test	23
10.5	Flakiness Index Test	23
10.6	Elongation Index Test	23
10.7	Aggregate Crushing Value (ACV) Test	24
10.8	Pycnometer Bottle Test (Specific Gravity and Water Absorption)	24
10.9	Vicat Apparatus Test	25
10.10	Le Chatelier Apparatus Test	25
10.11	Hydrometer Analysis	25
10.12	Slump Test	26
10.13	Bitumen Tests	26
10.13.1	Penetration Test	26
10.13.2	Softening Point Test (Ring and Ball Method)	26
10.13.3	Ductility Test	27
10.13.4	Specific Gravity Test	27
10.13.5	Flash and Fire Point Test	27
11	Batching Plant and Bitumen Mixing Plant	28
11.1	Batching Plant Operations	28
11.2	Asphalt/Bitumen Mixing Plant	28
11.3	Material Handling and Storage	28
11.4	Quality Control in Plant Operations	28
12	Environmental and Sustainability Practices	32
12.1	Waste Management	32
12.2	Dust and Noise Control	32
12.3	Water Management at Site	32
13	Learning Outcomes and Skills Gained	32
13.1	Technical Skills	32
13.2	Practical Exposure	32
13.3	Professional Development	32
14	Codal Provisions	33
14.1	Indian Standards (IS Codes)	33
14.2	IRC Codes	33
14.3	MoRTH Specifications	33
15	Conclusion	34

1 Introduction

Industrial training is an essential part of engineering education as it bridges the gap between theoretical knowledge and practical application. This report is based on a one-month winter internship carried out from 22nd December 2025 to 21st January 2026 under the National Highways Authority of India (NHAI) as the client and Rail Vikas Nigam Limited (RVNL) as the contractor.

The project titled “Six Lane Elevated Kona Expressway from KM 0.145 to KM 7.337 of NH-117 in the State of West Bengal” is approximately 7.19 km long and aims to improve traffic flow and connectivity in the Kona region. During the internship, I gained exposure to pile foundation works, PSC girder casting, superstructure erection, concrete and bitumen testing, and batching plant operations.

2 Objectives of the Internship

- To gain practical exposure to highway and bridge construction practices, including pile foundations, PSC girder casting, and superstructure works.
- To understand quality control procedures through material testing of concrete, bitumen, and aggregates at site and laboratory levels.
- To develop professional skills in site management, safety practices, and coordination within large infrastructure projects.

3 Description about the Organization

The National Highways Authority of India (NHAI) is a statutory body under the Ministry of Road Transport and Highways, Government of India, responsible for the development, maintenance, and management of national highways across the country.

Rail Vikas Nigam Limited (RVNL), a Government of India enterprise, served as the main contractor for this project and is involved in executing large infrastructure works such as highways, railways, bridges, and metro systems.

The project construction activities were carried out by the GPT Group, a leading infrastructure company known for its expertise in highways, bridges, and large-scale civil engineering projects, ensuring quality execution and timely completion of works.

4 Project Description

4.1 Project Location and Scope

This project is located along the designated highway corridor and involves the construction of a multi-lane elevated expressway to improve traffic flow and regional connectivity. The scope of the project includes roadway development, structural works, drainage systems, and associated infrastructure facilities.

4.2 Alignment, Length, and Structural Components

The alignment follows the approved highway corridor with minimal land acquisition and optimized geometric design. The total project length covers the specified stretch, incorporating major structural

components such as viaducts, piers, abutments, girders, deck slabs, and expansion joints to ensure structural integrity and serviceability.

4.3 Construction Methodology

The construction methodology involves systematic execution of foundation works, substructure construction, superstructure erection, and deck slab casting. Modern construction techniques such as precast prestressed girders, mechanized launching, quality-controlled concrete pouring, and staged traffic management are adopted to ensure safety, efficiency, and timely project completion.

5 Casting Yard Operations

5.1 Layout and Function of Casting Yard

The casting yard is systematically laid out to accommodate six separate casting beds for parallel girder production. Each casting bed is approximately 120 m in length and is equipped with stressing abutments, anchorage systems, rail tracks for trolley movement, and modular formwork arrangements. The yard also includes designated areas for material storage, batching plant access, curing zones, and stacking yards to ensure smooth workflow, quality control, and efficient handling of precast components.



5.2 Materials Used (Cement, Aggregates, Steel, Admixtures)

Ordinary Portland Cement (OPC), well-graded coarse and fine aggregates, high-tensile prestressing steel strands, and approved chemical admixtures are used in the casting process. Superplasticizers are added to improve workability while maintaining the required strength. All materials are tested and approved as per relevant IS codes before use to ensure durability, performance, and consistency in production.

5.3 Pre-tensioning and Post-tensioning System

The prestressing system adopted in this project includes both pre-tensioning and post-tensioning methods. **Pre-tensioning is predominantly used in the casting yard for girder production due to its suitability for mass production, better quality control, and higher structural efficiency.** In this method, high-strength steel strands are tensioned between fixed abutments before concrete casting,

and after the concrete attains sufficient strength, the prestress is transferred to the concrete through bond action upon release of the strands.

In post-tensioning, ducts or de-bonding pipes are placed within the concrete to allow stressing after hardening. This method is mainly adopted where longer spans, segmental construction, or on-site prestressing is required. **Post-tensioning allows for controlled stressing, reduced cracking, and improved load-carrying capacity in in-situ structural elements.**

Different diameters of prestressing cables are used, with **larger diameter strands placed at the bottom to resist tensile stresses due to bending, and three cables placed near the top to control cracking, balance stresses, and enhance overall structural performance.** The use of de-bonding pipes ensures proper load transfer, corrosion protection, and effective force distribution along the tendon length.



(a) Use jack for creating pretension



(b) Pretensile cable is used in different dia

5.4 PSC Girder Casting Process

The PSC girder casting process begins with proper shuttering and precise alignment of formwork, followed by reinforcement cage fabrication and inspection. Cover blocks are provided at reinforcement locations to maintain the specified concrete cover, ensuring corrosion protection, fire resistance, and proper load transfer. Prestressing cables are then placed and stressed to the specified force using hydraulic jacks, and elongation values are measured and verified.

De-bonding pipes are provided near the girder ends to prevent bond between the concrete and tendons, thereby reducing stress concentration and minimizing end cracking. Concrete is poured in a controlled manner and compacted using mechanical vibrators to avoid honeycombing and ensure uniform density. Slump tests and cube strength tests are conducted regularly to monitor concrete quality and compliance.

5.5 Curing, De-shuttering, and Storage of Girders

Curing: Curing is a crucial process carried out after casting to ensure proper hydration of cement, development of design strength, and long-term durability of the concrete. In the casting yard, girders are cured for a period of 20 to 28 days using continuous water curing, wet coverings such as hessian cloth, or approved curing compounds. Proper curing prevents rapid moisture loss, minimizes shrinkage and surface cracking, and enhances the mechanical properties and impermeability of concrete. Adequate curing also improves resistance to environmental effects, thereby increasing the service life of the prestressed concrete girders.

De-shuttering: De-shuttering involves the systematic and careful removal of formwork once the concrete has attained sufficient early strength to support its own weight and imposed construction loads. This operation is carried out in a controlled manner to avoid sudden load release, surface damage, edge chipping, or dimensional distortion of the girder. Special attention is given to corners, edges, prestressing zones, and haunched regions. Proper de-shuttering ensures dimensional accuracy, preserves surface finish, and prepares the girder for safe de-stressing, handling, and storage.

Storage of Girders: After de-shuttering and de-stressing, the girders are lifted using suitable cranes and transported to designated storage yards within the casting yard premises. The girders are placed on firm, level, and well-aligned supports at predetermined bearing points to prevent bending, torsion, or differential settlement. Timber or rubber pads are often used between the girder and supports to distribute load evenly and avoid local stress concentration. Adequate spacing is maintained between adjacent girders to allow inspection, ventilation, and safe movement of equipment. Protective measures such as covering, drainage arrangements, and regular inspection are provided to prevent damage due to weather exposure, handling, or accidental impact, ensuring that the girders remain structurally sound and ready for transportation and erection at the project site.



(a) Deshuttering



(b) vibrator after concrete pouring

6 Pre-stressed Concrete (PSC) Girders

Pre-stressed Concrete (PSC) girders are structural members in which internal compressive stresses are intentionally introduced to counteract the tensile stresses developed under service loads. This prestressing significantly enhances load-carrying capacity, span length, crack resistance, and durability compared to conventional reinforced concrete (RCC) girders. PSC girders are widely used in bridges, flyovers, and elevated corridors due to their efficiency, economy, and long-term performance.

subsection Use of PSC Girders and I-Section Configuration Prestressed concrete (PSC) girders are preferred over reinforced concrete (RCC) girders due to their higher load-carrying capacity, longer span capability, reduced structural depth, better crack control, and improved durability.

The I-section configuration is adopted because it provides maximum flexural strength with minimum material by concentrating concrete in the flanges and reducing weight in the web. The bottom flange

(lower width) of the girder is 1020 mm, while the top flange (upper width) is 1100 mm, with an allowable dimensional tolerance of $\pm 3\%$. This shape enhances structural efficiency, ease of handling, and compatibility with deck slab construction.

A total of six girders are produced for each span, consisting of four large girders of 40 m span and two smaller girders of 34 m span. All girders have uniform spacing and standardized reinforcement detailing for consistency and quality control.



(a) I Section Girder

6.1 Cable Layout and Anchorage System

The prestressing cable layout is designed to counteract the bending moments and shear forces experienced by the girder. High-strength steel strands are arranged in a parabolic profile, with a greater concentration of cables near the bottom flange at mid-span where tensile stresses are maximum, and gradually rising toward the top flange near the supports to counteract negative bending moments.

Anchorage systems are installed at the stressing ends of the girders to securely grip and transfer the prestressing force from the steel strands to the concrete. These systems consist of anchor heads, wedges, bearing plates, and anchorage blocks designed to resist high tensile forces without slippage or failure. De-bonding pipes are often provided near the girder ends to reduce stress concentration and minimize cracking in anchorage zones.

6.2 Stress-Strain Behavior in Upward and Downward Loading Cases

The stress-strain behavior of PSC girders differs significantly from that of RCC girders due to the presence of initial compressive prestress. Under downward loading (service loads such as self-weight, superimposed dead load, and live load), tensile stresses develop at the bottom fibers of the girder. However, the prestressing force introduces compressive stresses in these regions, effectively reducing or eliminating tension, thereby preventing cracking and increasing stiffness.

Under upward loading conditions (such as lifting during handling, transportation, or erection), tensile stresses may develop at the top fibers of the girder. The presence of top prestressing cables and appropriate section geometry ensures that these tensile stresses remain within permissible limits. The I-section configuration plays a crucial role in both cases by providing greater moment of inertia, improved stress distribution, and enhanced resistance to bending in both positive and negative moment conditions.

6.3 Quality Control Measures

Strict quality control measures are implemented throughout the fabrication of PSC girders to ensure structural safety, durability, and compliance with design specifications. These measures include:

- Inspection and approval of raw materials such as cement, aggregates, prestressing steel, and admixtures as per relevant IS codes.
- Verification of reinforcement detailing, cable layout, and anchorage positioning before casting.
- Calibration and inspection of stressing equipment, including hydraulic jacks and pressure gauges.
- Monitoring of concrete quality through slump tests, cube strength tests, and curing records.
- Measurement and verification of prestressing force and elongation values during stressing and de-stressing operations.
- Visual inspection of girders after de-shuttering and de-stressing for cracks, surface defects, dimensional accuracy, and alignment.

These quality control practices ensure consistent girder performance, minimize construction defects, and enhance the overall reliability and service life of the bridge structure.

6.4 Stressing and De-stressing Operations and Equipment

Stressing and de-stressing are critical stages in the prestressed concrete (PSC) girder casting process, as they introduce and subsequently transfer compressive forces into the concrete to counteract tensile stresses developed under service loads. At the casting yard, stressing is carried out using hydraulic power packs connected to both large six-jack systems and smaller three-jack systems, depending on the girder size and cable configuration. Oil is used as the hydraulic fluid to transmit pressure efficiently and ensure smooth jack operation.

High-strength prestressing strands are anchored at the stressing end blocks and tensioned to the specified force using calibrated hydraulic jacks. For 40 m span girders, a stress level of approximately 50 kg/cm^2 is applied to each cable during stressing. The applied force is carefully controlled, and elongation measurements are recorded and compared with theoretical values to verify the accuracy of stressing and to detect any losses due to friction or seating. Stressing is generally performed after the concrete has attained sufficient early strength, typically between 3 to 7 days after casting. Once the required strength is achieved, de-stressing is carried out to safely transfer the prestressing force from the steel strands to the concrete. The de-stressing process is performed gradually and symmetrically to avoid shock loading, uneven stress distribution, or cracking of the girder. During de-stressing, the anchorage barrel retracts by approximately 450 mm, indicating proper and controlled transfer of prestress.

Strict safety protocols are followed during both stressing and de-stressing operations, including restricted access to the stressing zone, proper alignment of jacks and anchorages, continuous monitoring of pressure gauges, and use of personal protective equipment. These measures ensure safe execution, minimize the risk of tendon failure, and guarantee uniform prestress application, thereby enhancing the structural performance, serviceability, and durability of the PSC girders.



(a) Stressing end block and cable arrangement at the casting yard.



(b) Hydraulic jacks and anchorage setup during prestressing operations

6.5 Stressing Calculations and Procedure

Stressing calculations form a critical part of prestressed concrete (PSC) girder construction, ensuring that the required prestressing force is accurately applied to achieve the desired structural performance, serviceability, and durability. The calculations are carried out in accordance with IS 1343:2012, IRC:112, and MoRTH specifications and are verified on site through elongation measurements and pressure gauge readings.

6.5.1 Design Stress and Jacking Force

From the site records, **12.7 mm diameter 7-wire low-relaxation strands** were used, each having a cross-sectional area of **140 mm²** and an ultimate tensile strength (UTS) of **1860 N/mm²**. The jacking stress was limited to approximately **75% of UTS**, i.e.,

$$f_{pj} = 0.75 \times 1860 = 1395 \text{ N/mm}^2$$

The corresponding jacking force per strand was:

$$P_j = A_p \times f_{pj} = 140 \times 1395 = 195,300 \text{ N} \approx 195.3 \text{ kN}$$

6.5.2 Elongation Calculation

The theoretical elongation was computed using:

$$\Delta L = \frac{P_j \times L}{A_p \times E_p}$$

where:

$$L = 25,000 \text{ mm (tendon length)}, \quad E_p = 2 \times 10^5 \text{ N/mm}^2$$

$$\Delta L = \frac{195,300 \times 25,000}{140 \times 2 \times 10^5} = 174.4 \text{ mm}$$

This value was used as the benchmark for field verification.

6.5.3 Site-Measured Elongation and Verification

Based on the site stressing sheets extracted from the ZIP file, the observed elongations closely matched the theoretical values and were within permissible limits.

Table 1: Stressing Observation Record (Actual Site Data)

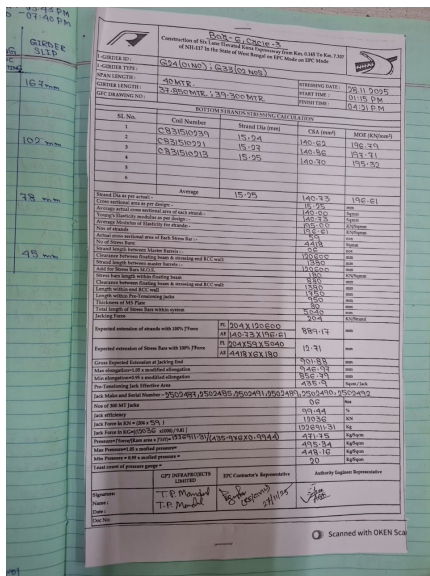
Tendon No.	Jack Pressure (MPa)	Theoretical Elongation (mm)	Measured Elongation (mm)	Difference
1	32.0	174	172	1.15
2	32.0	174	176	1.15
3	32.0	174	173	0.57
4	32.0	174	175	0.57

6.5.4 Permissible Tolerances

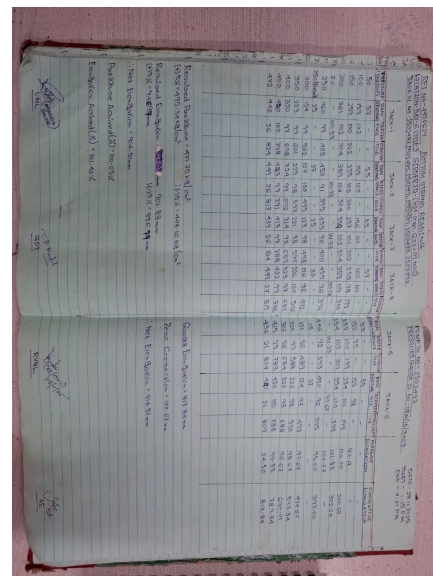
As per IS 1343 and IRC provisions:

$$\pm 7\% \text{ (for pre-tensioning)} \quad \text{and} \quad \pm 5\% \text{ (for post-tensioning)}$$

were considered acceptable. All observed values remained well within these limits, confirming the accuracy of the stressing operation.



(a) calculation chart 1



(b) calculation chart 2

6.5.5 Stressing Sequence and Safety Measures

Stressing was carried out in a controlled and systematic sequence to ensure uniform force distribution across all strands. Hydraulic jacks and pressure gauges were calibrated prior to operation. Proper anchorage seating, gradual loading, and wedge locking were ensured. Safety measures such as exclusion zones, use of personal protective equipment (PPE), and controlled release of strands were strictly enforced.

6.6 Haunching and Reinforcement Detailing

Haunching is provided near supports and girder-deck junctions to increase effective depth and improve stress distribution. Haunch sizes typically vary between 200 mm, 150 mm, and 100 mm.

For reinforcement detailing, 10 mm diameter bars are used within the haunch zones for shear resistance and crack control. Additionally, at both ends of each girder, seven numbers of 20 mm diameter bars are provided to strengthen the anchorage zones and safely resist prestressing forces.



6.7 Girder Erection Operations

Girder erection is a critical construction activity requiring detailed planning, strict safety control, and coordinated execution. The process begins with clear identification of the **Entry Point** and **Exit Point** for all personnel and equipment. An **Assembly Point** is designated for emergency situations, and the **Emergency Planning Programme (EPP)** is implemented with provisions such as ambulance availability. Site emergency contact numbers are displayed prominently, and the **Safety Expert: SK Inayat (Ph: 7063219336)** is responsible for overall supervision and safety compliance.

Before erection, **Hazard Identification and Risk Assessment (HIRA)** is conducted on an activity-wise basis to evaluate site-specific risks and implement appropriate control measures. The erection process is carried out in two main stages: **Planning** and **Execution**. All lifting equipment, including cranes, slings, and hooks, undergoes **Third Party Inspection (TPI)** to verify certification, load capacity, and operational readiness. A valid permit-to-work system and maintenance checklist are ensured prior to commencement.

Erection is performed using **two cranes in tandem lifting configuration** or a **launcher** for single-span lifts, depending on site conditions. The girder weighs approximately **110 tonnes**, with an additional **3 tonnes** considered for lifting accessories, resulting in a total lifting load of **113 tonnes**. When two cranes are used, the load is equally shared, and the load per crane is **56.5 tonnes**. Given the crane Safe Working Load (SWL) of **72 tonnes**, the safety utilization percentage is calculated as:

$$\text{Safety Utilization} = \frac{56.5 \times 100}{72} = 78.5\%$$

This satisfies the safety requirement of maintaining at least **20% reserve capacity**, thereby confirming that the lifting operation is within safe limits.

During erection, the girder is stabilized using **turnbuckles**, **side supports**, and **cross supports** to prevent lateral movement, torsional instability, and misalignment. Continuous monitoring is carried out by the safety team and erection supervisors to ensure adherence to safety protocols and alignment tolerances throughout the lifting and placement process.

6.7.1 Risk Control Matrix for Girder Erection (HIRA)

Table 2: Risk Control Matrix for Girder Erection Operations

Sl. No.	Hazard	Potential Risk	Risk Level	Control Measures
1	Crane overloading	Crane failure, collapse	High	SWL verification, load calculation, dual crane sharing, TPI certification
2	Improper rigging	Sling failure, load drop	High	Certified riggers, inspection of slings, use of correct lifting points
3	Wind and weather conditions	Loss of control, swinging of girder	Medium	Monitor wind speed, suspend operation during adverse weather
4	Personnel in lifting zone	Injury or fatality	High	Barricading, restricted access, safety signage, spotters
5	Poor ground bearing capacity	Crane overturning	High	Ground compaction, use of crane mats and load distribution plates
6	Misalignment during placement	Structural damage, rework	Medium	Use of guide ropes, survey checks, slow positioning
7	Communication failure	Unsafe operation	Medium	Dedicated signaller, standard hand signals, radio communication
8	Equipment failure	Accidents and delays	High	Preventive maintenance, daily equipment inspection
9	Working at height	Fall hazards	High	Use of safety harness, lifelines, guardrails
10	Emergency response unpreparedness	Delayed rescue, increased severity	Medium	Emergency planning programme (EPP), drills, ambulance availability

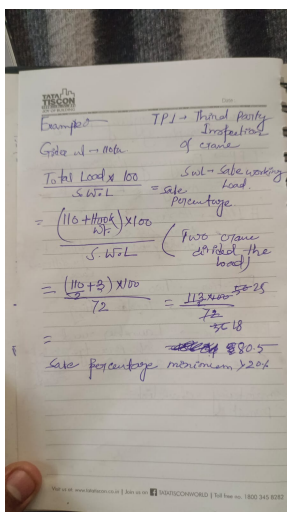
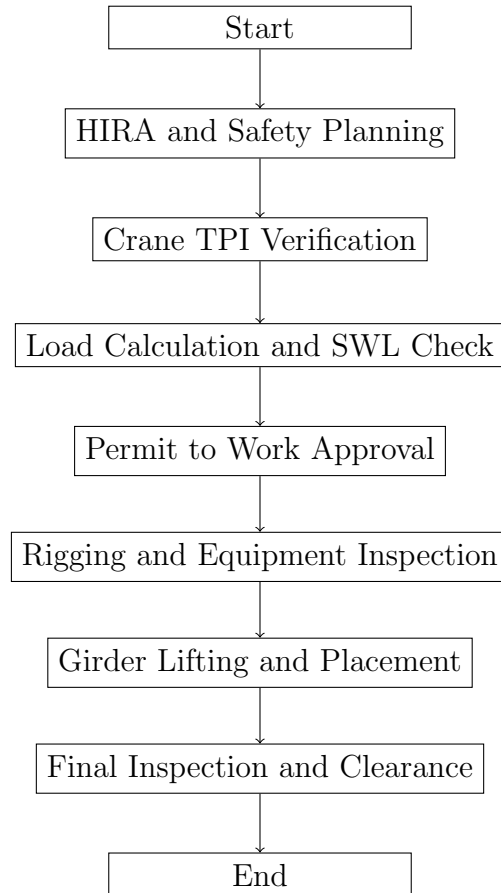


Table 3: Crane and Lifting Equipment Safety Checklist

Sl. No.	Checklist Item	Status	Remarks
1	Valid Third Party Inspection (TPI) certificate available	Yes	Verified and valid
2	Crane SWL clearly marked and verified	Yes	SWL within safe limits
3	Load calculation sheet approved	Yes	Approved by engineer
4	Hydraulic system, brakes, and controls functional	Yes	Checked and operational
5	Slings, shackles, hooks inspected and certified	Yes	Certified lifting accessories
6	Ground bearing capacity and crane mats adequate	Yes	Mats provided and stable
7	Wind speed within permissible limits	Yes	Weather conditions acceptable
8	Communication system (radio/hand signals) established	Yes	Signalman appointed
9	Barricading and exclusion zone provided	Yes	Area properly secured
10	Emergency response arrangements in place	Yes	Ambulance and EPP available
11	Permit to work approved and displayed	Yes	Permit issued and displayed
12	Safety briefing and toolbox talk conducted	Yes	Toolbox talk completed

6.7.2 Girder Erection Procedure Flowchart



7 Pile Foundation Works

7.1 Types of Piles Observed

Bored cast-in-situ reinforced concrete piles were primarily observed at the site. These piles are suitable for deep foundations where surface soil strata are weak and heavy structural loads must be transferred to deeper, stronger soil layers. The piles function either as **friction piles**, **end-bearing piles**, or a combination of both, depending on the soil stratification and design requirements. Both **under-reamed piles** and **straight-shaft piles** were observed in different locations to enhance load capacity and settlement performance.

7.2 Piling Equipment and Methods

Hydraulic rotary piling rigs, crawler cranes, tremie pipes, reinforcement cages, and slurry circulation systems were used for pile construction. A **hydraulic rotary rig** was employed for drilling, capable of boring through dense strata and mixed soil profiles. Polymer slurry was adopted instead of bentonite to stabilize boreholes during drilling and concreting operations due to its lower residue, ease of removal, and improved bond between concrete and soil.

7.3 Boring, Concreting, and Reinforcement

Boring was carried out using rotary drilling techniques. Polymer slurry, prepared using sodium lignosulphonate-based compounds, was circulated to maintain bore stability and prevent collapse. The polymer slurry had a viscosity range of **60 to 120 seconds** (Marsh cone) and a pH value between **10 and 12**, ensuring effective suspension of soil particles.



After reaching the required depth, the bore was cleaned using air-lift or flushing methods. Prefabricated reinforcement cages were then lowered into the boreholes with proper centralizers to maintain cover. Concreting was carried out using **tremie pipes**, ensuring continuous, segregation-free placement from the bottom upwards. The tremie pipe was always kept embedded at least **1.5 to 2.0 m** inside fresh concrete to avoid slurry contamination.

7.4 Pile Load Tests and Integrity Tests

Pile quality and performance were verified through a series of field tests, including:

- **Standard Penetration Test (SPT)** for soil investigation and stratification assessment,

- **Initial Pile Load Tests** to confirm design load capacity,
- **Routine Pile Load Tests** for quality control of production piles, and
- **Pile Integrity Tests (PIT)** to detect defects such as voids, necking, or discontinuities in the pile shaft.

These tests ensured compliance with IS codes and design requirements.

7.5 Standard Penetration Test (SPT): Procedure, Requirements, and Interpretation

The Standard Penetration Test (SPT) was conducted at the site to evaluate soil strength, density, and bearing characteristics, particularly for foundation design and pile capacity assessment.

7.5.1 Purpose of SPT

The SPT provides:

- Soil stratification details,
- Relative density of cohesionless soils,
- Consistency of cohesive soils,
- Estimation of bearing capacity, and
- Input parameters for foundation and pile design.

7.5.2 Equipment Used

The following standard equipment was used:

- Split-spoon sampler (outer diameter: 50.8 mm, inner diameter: 35 mm),
- Standard hammer weighing **63.5 kg**,
- Free fall height of **750 mm**,
- Drill rods,
- Anvil and drive head,
- Tripod or drilling rig with lifting arrangement.

7.5.3 SPT Procedure

1. Borehole drilling was carried out to the desired test depth using rotary or wash boring methods.
2. Loose soil at the bottom of the borehole was cleaned.
3. The split-spoon sampler was lowered to the borehole bottom.

4. The sampler was driven into the soil using a **63.5 kg hammer** falling freely from a height of **750 mm**.
5. The sampler penetration was recorded for three successive increments of **150 mm** each, totaling **450 mm**.
6. The number of blows required for the first 150 mm penetration was recorded but ignored.
7. The blows required for the second and third 150 mm penetrations were added to obtain the **SPT N-value**.



(a) Polymer mixing



7.5.4 Corrections Applied to N-Value

The field-measured N-value was corrected for:

- Overburden pressure correction,
- Hammer energy efficiency correction,
- Borehole diameter correction,
- Rod length correction,
- Sampler correction (with or without liners).

The corrected value is denoted as:

$$N_{60} = N \times C_E \times C_B \times C_R \times C_S$$

where:

- C_E = Hammer energy correction,
- C_B = Borehole diameter correction,
- C_R = Rod length correction,
- C_S = Sampler correction.

Table 4: Interpretation of SPT N-Values

N-Value	Soil Density/Consistency	Engineering Description
0–4	Very loose / Very soft	Low bearing capacity
4–10	Loose / Soft	Low to moderate strength
10–30	Medium dense / Medium stiff	Suitable for shallow foundations
30–50	Dense / Stiff	Good bearing capacity
>50	Very dense / Very stiff	Excellent bearing stratum

7.5.5 Interpretation of SPT N-Value

7.5.6 Field Observations from Site Notes

From the site sketches and notes:

- The SPT N-values reached approximately **60** at deeper strata, indicating very dense soil or hard strata suitable for end-bearing piles.
- The pile foundations included both **friction piles** and **end-bearing piles**, depending on depth and soil resistance.
- The pile cap, pier, pier cap, bearing pedestal, girders, deck slab, and crash barrier components were clearly identified and integrated into the foundation system.
- A **hydraulic rotary rig** was used for boring, confirming suitability for dense and mixed soil profiles.

7.5.7 Importance of SPT in Pile Design

The SPT results were used to:

- Determine pile length and founding depth,
- Select pile type (friction, end-bearing, or combined),
- Estimate ultimate and safe bearing capacities,
- Verify soil improvement requirements, and
- Support design validation through correlation with pile load tests.

8 Reinforcement Detailing of Pile

8.1 General Description

The bored cast-in-situ reinforced concrete piles were reinforced with a centrally placed longitudinal reinforcement cage and lateral ties (spiral or circular links) to ensure structural integrity, load transfer, and crack control. The reinforcement detailing was designed as per relevant Indian Standards and NHAI specifications to withstand axial loads, bending moments, and shear forces.

8.2 Pile Geometry and Size

- **Pile Diameter:** As per drawing (typically 1200 mm or as specified)
- **Pile Length:** As per founding level and soil profile
- **Concrete Grade:** M35 / M40 (as per project specification)
- **Clear Cover:** 75 mm (minimum) for durability in aggressive environment

8.3 Longitudinal Reinforcement

The pile was reinforced with uniformly distributed high-yield deformed bars arranged in a circular pattern.

- **Main Bars:** 16 to 20 numbers of **25 mm diameter** HYSD bars (as per drawing)
- **Arrangement:** Equally spaced around the circumference
- **Anchorage:** Bars were extended into the pile cap with adequate development length
- **Lap Length:** Provided where required, as per IS 456 and IS 2911

8.4 Lateral Reinforcement (Ties / Helical Reinforcement)

To prevent buckling of longitudinal bars and to confine concrete, lateral reinforcement was provided in the form of circular ties or helical spirals.

- **Tie Diameter:** 8 mm or 10 mm mild steel bars
- **Spacing:**
 - 100 mm c/c in top and bottom 1.5D zone of the pile
 - 150 mm to 200 mm c/c in the central portion of the pile shaft
- **Configuration:** Closed circular links with proper hooks and laps

8.5 Pile Head Reinforcement and Connection with Pile Cap

The pile head was specially reinforced to ensure effective load transfer and to resist punching and splitting forces at the pile–pile cap interface.

- Additional confinement reinforcement was provided in the pile head zone.
- Longitudinal bars were projected into the pile cap and anchored properly.
- Extra stirrups were provided near the pile head for enhanced shear resistance.

8.6 Corrosion Protection and Durability Measures

- Adequate concrete cover of 75 mm was maintained using cover blocks.
- Polymer slurry was fully removed before concreting to ensure proper bond.
- Concrete was placed using tremie pipes to avoid segregation and contamination.

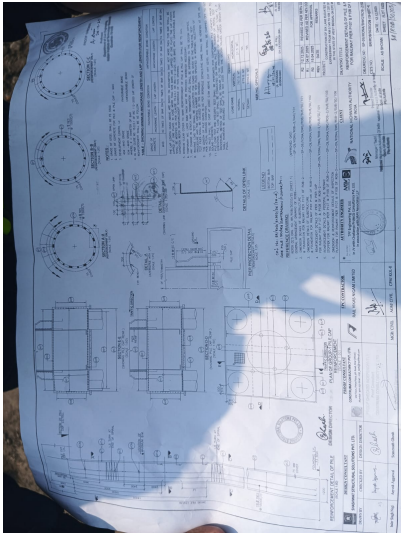
8.7 Reinforcement Summary Table

Table 5: Pile Reinforcement Details

Component	Specification	Remarks
Pile Diameter	As per drawing (e.g., 1200 mm)	Bored cast-in-situ pile
Concrete Grade	M35 / M40	As per design requirement
Main Bars	16–20 bars of 25 mm dia HYSD	Uniformly spaced
Lateral Ties	8–10 mm dia circular links	Confinement reinforcement
Tie Spacing	100 mm (ends), 150–200 mm (middle)	As per code and drawing
Clear Cover	75 mm	Durability requirement
Pile Head Bars	Extended into pile cap	Adequate anchorage

8.8 Construction and Placement Notes

- Reinforcement cages were fabricated on the ground and lifted using cranes.
- Centralizers were fixed at regular intervals to maintain uniform cover.
- Cages were lowered carefully into the boreholes without distortion.
- Concreting was carried out immediately after cage placement using tremie method.



(a) Drawing



(b) Excavation Work

9 Superstructure Construction

9.1 Pier and Pier Cap Construction

Piers were constructed to safely transfer the loads from the superstructure to the foundation system. The construction process involved layout marking, excavation (where required), reinforcement detailing as per approved drawings, and fabrication of reinforcement cages. These cages were erected using cranes and positioned accurately within the formwork. High-strength concrete was placed using transit mixers and concrete pumps, ensuring adequate compaction through mechanical vibrators. After achieving the

required strength, formwork was removed and curing was carried out using water spraying and curing compounds. Subsequently, pier caps were cast over the completed piers to distribute loads from multiple bearings and girders uniformly, ensuring proper alignment, level control, and structural continuity.

9.2 Bearing Installation

Bearings were installed on the pier caps to facilitate controlled movements of the superstructure due to thermal expansion, shrinkage, creep, traffic loading, and seismic forces. Prior to installation, bearing pedestals were checked for line, level, and surface finish. Non-shrink grout was used to level the bearing seats and ensure full contact. Elastomeric bearings were then positioned accurately as per orientation markings and design specifications. Proper alignment and fixing were verified before girder erection to ensure effective load transfer and long-term performance of the structure.

9.3 Deck Slab and Wearing Course

The deck slab was constructed over the girders to form the main riding surface of the bridge. Formwork and staging were erected to support the slab during concreting, and reinforcement was placed as per structural drawings, ensuring adequate cover and continuity. Concrete was poured in a controlled sequence to avoid cold joints and was compacted using needle vibrators. After completion, the slab was cured for the specified duration to achieve the desired strength and durability. Subsequently, a wearing course was laid over the deck slab to provide a smooth riding surface, enhance skid resistance, protect the structural concrete from abrasion and environmental exposure, and improve the overall service life of the bridge.

9.4 Expansion Joints and Drainage Provisions

Expansion joints were provided at appropriate locations to accommodate longitudinal movements of the superstructure and prevent distress due to temperature variations and shrinkage. Drainage spouts and scuppers were installed along the deck to ensure efficient removal of surface water, thereby preventing water stagnation, corrosion of reinforcement, and deterioration of structural components.

10 Laboratory Testing

Laboratory testing ensures the quality, strength, durability, and performance of materials used in highway and bridge construction. All tests were conducted in accordance with relevant Indian Standards (IS) to ensure compliance with design specifications and safety requirements.

10.1 Concrete Cube Compressive Strength Test

Relevant IS Code: IS 516:1959

Objective: To determine the compressive strength of hardened concrete.

Apparatus: Compression Testing Machine (CTM), cube moulds (150 mm × 150 mm × 150 mm), curing tank, weighing balance.

Procedure:

- Fresh concrete is placed into moulds in three layers and each layer is compacted using a high-frequency vibrator.

- The top surface is finished and cubes are stored for 24 hours.
- Specimens are demoulded and cured in clean water at $27 \pm 2^\circ\text{C}$ for 7, 14, or 28 days.
- The cube is wiped dry and placed centrally in the CTM.
- Load is applied uniformly at a rate of approximately $140 \text{ kg/cm}^2/\text{min}$ until failure.
- The maximum load at failure is recorded.

Calculation:

$$\text{Compressive Strength (MPa)} = \frac{\text{Maximum Load (N)}}{\text{Loaded Area (mm}^2\text{)}}$$

Acceptance Criteria: The average strength of three cubes shall not be less than the specified grade of concrete (e.g., M30, M35).

Significance: Confirms the load-bearing capacity and structural safety of concrete.

10.2 High-Frequency Motor Concrete Cube Vibrator

Relevant IS Code: IS 516:1959

Objective: To achieve proper compaction of concrete specimens by eliminating entrapped air.

Description: A motor-driven vibrating table or needle vibrator is used to ensure uniform density and prevent honeycombing in concrete cubes.

Procedure:

- Concrete is filled into moulds in layers.
- Each layer is vibrated until air bubbles stop rising.
- Excess concrete is struck off and surface finished smoothly.

Significance: Proper compaction improves strength, durability, and accuracy of test results.

10.3 Aggregate Impact Value (AIV) Test

Relevant IS Code: IS 2386 (Part IV):1963

Objective: To assess the toughness of aggregates and their resistance to sudden impact loads.

Apparatus: Aggregate impact testing machine, cylindrical metal measure, tamping rod, 2.36 mm sieve, oven, weighing balance.

Procedure:

- The aggregate sample is oven-dried and sieved to obtain the required size fraction.
- The sample is filled into the mould in three layers, each layer tamped 25 times.
- The mould is placed in the testing machine and subjected to 15 blows of a standard hammer dropped from a height of 380 mm.
- The crushed sample is sieved through a 2.36 mm sieve and the fines are weighed.

Calculation:

$$\text{Aggregate Impact Value (AIV)} = \frac{\text{Weight of fines}}{\text{Total weight of sample}} \times 100$$

Acceptance Criteria: For wearing surfaces, AIV should not exceed 30%; for non-wearing surfaces, 45%.

Significance: Indicates the suitability of aggregates for pavements and high-stress applications.

10.4 Concrete Permeability Test

Relevant IS Code: IS 3085:1965

Objective: To determine the permeability of concrete to water under pressure.

Apparatus: Permeability test cell, water pressure pump, concrete cube specimens, measuring scale.

Procedure:

- A cured concrete cube is placed in the permeability cell and sealed properly.
- Water pressure is applied gradually and maintained for a specified duration.
- After completion, the specimen is split open and the depth of water penetration is measured.

Significance: Low permeability indicates better durability, resistance to corrosion, and improved service life.

10.5 Flakiness Index Test

Relevant IS Code: IS 2386 (Part I):1963

Objective: To determine the percentage of flaky particles in aggregate.

Apparatus: Flakiness gauge, standard sieves, oven, weighing balance.

Procedure:

- Aggregates are sieved into different size fractions.
- Each fraction is tested using the flakiness gauge.
- Particles passing through the slot are classified as flaky and weighed.

Calculation:

$$\text{Flakiness Index} = \frac{\text{Weight of flaky particles}}{\text{Total weight of sample}} \times 100$$

Acceptance Criteria: Generally, the flakiness index should not exceed 25–30% for concrete and pavement works.

Significance: Excess flaky particles reduce strength, workability, and durability of concrete.

10.6 Elongation Index Test

Relevant IS Code: IS 2386 (Part I):1963

Objective: To determine the percentage of elongated particles in aggregate.

Apparatus: Elongation gauge, standard sieves, weighing balance.

Procedure:

- Aggregates are sieved into size fractions.
- Each fraction is tested using the elongation gauge.
- Particles retained on the gauge are considered elongated and weighed.

Calculation:

$$\text{Elongation Index} = \frac{\text{Weight of elongated particles}}{\text{Total weight of sample}} \times 100$$

Acceptance Criteria: Elongation index is generally limited to 25–30%.

Significance: High elongation index leads to poor packing and reduced strength.

10.7 Aggregate Crushing Value (ACV) Test

Relevant IS Code: IS 2386 (Part IV):1963

Objective: To determine the resistance of aggregates to crushing under compressive load.

Apparatus: Compression testing machine, cylindrical mould, plunger, standard sieves.

Procedure:

- Aggregate sample is filled into the mould and compacted.
- A compressive load of 40 tonnes is applied gradually.
- The crushed material is sieved and fines are weighed.

Calculation:

$$\text{Aggregate Crushing Value (ACV)} = \frac{\text{Weight of fines}}{\text{Total weight of sample}} \times 100$$

Acceptance Criteria: ACV should generally be less than 30% for wearing courses and less than 45% for base courses.

Significance: Indicates the strength and durability of aggregates.

10.8 Pycnometer Bottle Test (Specific Gravity and Water Absorption)

Relevant IS Code: IS 2386 (Part III):1963

Objective: To determine the specific gravity and water absorption of fine aggregates.

Apparatus: Pycnometer bottle, oven, weighing balance, water bath.

Procedure:

- A sample of aggregate is oven-dried and weighed.
- The sample is placed in the pycnometer filled with water and weighed.
- The sample is then weighed in saturated surface dry (SSD) condition and submerged condition.

Significance: Specific gravity is essential for mix design calculations and quality control.

10.9 Vicat Apparatus Test

Relevant IS Code: IS 4031 (Part 4 & 5):1988

Objective: To determine the standard consistency, initial setting time, and final setting time of cement.

Apparatus: Vicat apparatus, plunger, initial and final setting needles, stopwatch.

Procedure:

- Cement paste is prepared and placed in the Vicat mould.
- Standard consistency is determined by plunger penetration.
- Initial and final setting times are determined using needles.

Acceptance Criteria: Initial setting time shall not be less than 30 minutes and final setting time shall not exceed 10 hours.

Significance: Ensures proper setting behavior for safe construction operations.

10.10 Le Chatelier Apparatus Test

Relevant IS Code: IS 4031 (Part 3):1988

Objective: To determine the soundness of cement.

Apparatus: Le Chatelier mould, water bath, boiling tank, Vernier caliper.

Procedure:

- Cement paste is placed in the mould and immersed in water.
- The mould is then boiled and cooled.
- Expansion is measured using calipers.

Acceptance Criteria: Expansion should not exceed 10 mm.

Significance: Prevents use of unsound cement that may cause cracks and failure.

10.11 Hydrometer Analysis

Relevant IS Code: IS 2720 (Part 4):1985

Objective: To determine particle size distribution of fine-grained soils.

Apparatus: Hydrometer, sedimentation cylinder, dispersing agent, thermometer.

Procedure:

- Soil is dispersed in water using a dispersing agent.
- Hydrometer readings are taken at specific time intervals.
- Particle sizes are calculated using Stokes' law.

Significance: Helps in soil classification and foundation design.

10.12 Slump Test

Relevant IS Code: IS 1199:1959

Objective: To determine the workability of fresh concrete.

Apparatus: Slump cone, tamping rod, base plate.

Procedure:

- Concrete is placed in the cone in three layers.
- Each layer is tamped 25 times.
- The cone is lifted vertically and slump is measured.

Acceptance Criteria: Slump value should conform to the specified range for the particular structural element.

Significance: Ensures ease of placement and compaction without segregation.

10.13 Bitumen Tests

10.13.1 Penetration Test

Relevant IS Code: IS 1203:1978

Objective: To determine the hardness or consistency of bitumen.

Apparatus: Penetrometer, standard needle, water bath, stopwatch.

Procedure:

- Bitumen sample is prepared and conditioned at 25°C.
- A standard needle is allowed to penetrate the sample under a load of 100 g for 5 seconds.
- The depth of penetration is measured in tenths of a millimeter.

Significance: Determines suitability of bitumen for different climatic conditions and traffic loads.

10.13.2 Softening Point Test (Ring and Ball Method)

Relevant IS Code: IS 1205:1978

Objective: To determine the temperature at which bitumen softens.

Apparatus: Ring and ball apparatus, thermometer, heating bath.

Procedure:

- Bitumen is filled in two rings and allowed to cool.
- Steel balls are placed on the samples.
- The assembly is heated at a uniform rate.
- The temperature at which the bitumen softens and touches the bottom plate is recorded.

Significance: Indicates temperature susceptibility and performance at high temperatures.

10.13.3 Ductility Test

Relevant IS Code: IS 1208:1978

Objective: To determine the ductility of bitumen.

Apparatus: Ductility testing machine, briquette moulds, water bath.

Procedure:

- Bitumen is cast into standard briquette moulds.
- After cooling, specimens are pulled apart at a uniform rate of 50 mm/min at 27°C.
- The elongation at the point of breaking is measured in centimeters.

Acceptance Criteria: Ductility value should generally not be less than 75 cm for paving grade bitumen.

Significance: Indicates the ability of bitumen to undergo deformation without cracking.

10.13.4 Specific Gravity Test

Relevant IS Code: IS 1202:1978

Objective: To determine the density and specific gravity of bitumen.

Apparatus: Pycnometer, water bath, weighing balance.

Procedure:

- Bitumen sample is weighed in air and water.
- Specific gravity is calculated from the recorded weights.

Significance: Used in mix design and quality control of bituminous mixtures.

10.13.5 Flash and Fire Point Test

Relevant IS Code: IS 1209:1978

Objective: To determine the temperatures at which bitumen vapors flash and ignite.

Apparatus: Pensky-Martens closed cup apparatus or Cleveland open cup apparatus.

Procedure:

- Bitumen sample is heated at a controlled rate.
- At regular temperature intervals, a test flame is introduced.
- The temperature at which a flash occurs is recorded as flash point.
- The temperature at which continuous burning occurs is recorded as fire point.

Acceptance Criteria: Flash point should generally be greater than 175°C for paving grade bitumen.

Significance: Ensures safety during heating, storage, and mixing operations.

11 Batching Plant and Bitumen Mixing Plant

11.1 Batching Plant Operations

The batching plant plays a vital role in ensuring the production of quality-controlled concrete by accurately proportioning cement, aggregates, water, and admixtures. At the project site, an automated batching plant with computerized control panels was used to maintain consistency in mix proportions. The plant followed the weigh batching method, ensuring precise measurement of materials as per approved mix design.

Aggregates were stored in separate bins based on size and fed into the plant through conveyor belts. Cement was stored in silos and conveyed pneumatically to the mixer. Admixtures were dosed through calibrated dispensers. Regular calibration of weighing systems and moisture correction of aggregates were carried out to maintain mix accuracy. Trial mixes were conducted, and concrete was produced only after approval from the quality control team.

11.2 Asphalt/Bitumen Mixing Plant

The asphalt mixing plant was used for the production of hot mix asphalt required for pavement layers such as Dense Bituminous Macadam (DBM) and Bituminous Concrete (BC). A hot mix plant with controlled temperature systems and automatic weighing mechanisms was employed to ensure uniform coating of aggregates with bitumen.

Aggregates were dried and heated in a rotating drum, while bitumen was heated separately in insulated storage tanks. The heated aggregates and bitumen were mixed in the pug mill or drum mixer as per job mix formula. Temperature control was strictly maintained during production, transportation, and laying to prevent premature setting and loss of workability.

11.3 Material Handling and Storage

Proper material handling and storage practices were followed to preserve material quality and prevent contamination. Aggregates were stored in designated stockpiles on hard, well-drained surfaces to avoid segregation and moisture variation. Cement was stored in moisture-proof silos and used on a first-in-first-out (FIFO) basis.

Bitumen was stored in insulated tanks equipped with temperature gauges and heating coils to maintain required viscosity. Admixtures and fillers were stored in covered areas with proper labeling. All materials were handled using loaders, conveyors, and pumping systems to minimize manual handling and ensure safety.

11.4 Quality Control in Plant Operations

Quality control measures were strictly implemented at both batching and asphalt plants to ensure compliance with project specifications and standards. Routine tests were conducted on incoming materials, including sieve analysis for aggregates, specific gravity tests, and bitumen penetration, ductility, and softening point tests.

For concrete, tests such as slump test, compaction factor test, and cube compressive strength test were conducted regularly. For asphalt mixes, Marshall stability and flow tests, bitumen content analysis,

and temperature checks were performed. Calibration of plant equipment, inspection of storage facilities, and verification of mix designs were part of daily quality assurance procedures to maintain consistent production quality.



(a) weight batching plant



(b) mixing plant



(a) Bitumen Mixing



(b) Bitumen and fuel tank



(a) In the RN Lab Kona



(a) Slump test



(b) Bitumen Test



(a) Different test apparatus



(b) Different test apparatus



(a) Permeability apparatus



(b) Cube test apparatus



(a) High Frequency Mortar Cube Vibrator



(b) Cube curing

12 Environmental and Sustainability Practices

12.1 Waste Management

Effective waste management practices were implemented at the construction site to minimize environmental impact. Construction waste such as concrete debris, steel scraps, packaging materials, and domestic waste was properly segregated at source. Reusable and recyclable materials were collected separately, while hazardous waste was safely stored and disposed of in accordance with environmental guidelines. These practices helped in maintaining site cleanliness and promoting sustainable construction.

12.2 Dust and Noise Control

Dust and noise control measures were strictly followed to ensure a safe and healthy working environment. Regular water sprinkling was carried out on haul roads, excavation areas, and material stockpiles to suppress dust. Construction materials were covered during transportation and storage. Noise pollution was controlled by maintaining machinery in good working condition, using silencers, and limiting the operation of heavy equipment during night hours or in sensitive zones near residential areas.

12.3 Water Management at Site

Water conservation and management were given high priority at the site. Water used for curing, washing aggregates, and dust suppression was reused wherever possible. Proper drainage systems were provided to prevent water stagnation and soil erosion. Monitoring of water consumption and controlled usage helped reduce wastage and supported sustainable construction practices.

13 Learning Outcomes and Skills Gained

13.1 Technical Skills

The internship significantly enhanced my technical knowledge of civil engineering construction practices, including pile foundation works, pre-stressed concrete girder casting, concrete mix design, and material testing procedures as per relevant IS codes. I also gained exposure to batching plant operations, bitumen mixing processes, and quality control methods.

13.2 Practical Exposure

The training provided valuable hands-on experience through direct involvement in site inspections, laboratory testing, and observation of real-time construction activities. This practical exposure helped me understand the sequence of construction operations, equipment usage, and the challenges faced during project execution.

13.3 Professional Development

The internship significantly improved my communication, teamwork, discipline, and time management skills. It also enhanced my understanding of workplace ethics, safety practices, and professional coordination, preparing me for future responsibilities in civil engineering.

14 Codal Provisions

All construction and testing activities were carried out in accordance with relevant Indian Standards (IS), Indian Roads Congress (IRC) codes, and MoRTH specifications.

14.1 Indian Standards (IS Codes)

- IS 456:2000 – Plain and Reinforced Concrete
- IS 10262:2019 – Concrete Mix Design
- IS 383:2016 – Aggregates for Concrete
- IS 516:1959 – Concrete Strength Tests
- IS 1199:1959 – Sampling and Testing of Concrete
- IS 4031 (Parts 1–15) – Cement Tests
- IS 2386 (Parts I–IV) – Aggregate Tests
- IS 73:2013 – Paving Bitumen
- IS 1201–1216 – Bitumen Tests
- IS 2720 (Relevant Parts) – Soil Tests
- IS 2911 (Parts 1–4) – Pile Foundation
- IS 1343:2012 – Prestressed Concrete
- IS 1786:2008 – Reinforcement Steel

14.2 IRC Codes

- IRC:6-2017 – Loads and Stresses for Road Bridges
- IRC:18-2000 – Prestressed Concrete Bridges
- IRC:21-2000 – RCC Bridges
- IRC:37-2018 – Flexible Pavement Design
- IRC:112-2020 – Concrete Road Bridges

14.3 MoRTH Specifications

- MoRTH Specifications for Road and Bridge Works (5th Revision)
- Section 1000 – Quality Control
- Section 1700 – Bituminous Works
- Section 2000 – Cement Concrete Pavement

15 Conclusion

The one-month winter internship under the National Highways Authority of India (NHAI), with project execution by GPT Group and construction supervision by Rail Vikas Nigam Limited (RVNL), provided comprehensive exposure to real-world highway and structural engineering practices. The training effectively bridged the gap between theoretical knowledge and practical implementation through direct involvement in planning, execution, quality control, and safety management of a large-scale infrastructure project.

Hands-on experience was gained in critical construction activities such as prestressed concrete (PSC) girder casting and erection, pre-tensioning and post-tensioning operations, pile foundation construction, and pile load and integrity testing. The internship also provided in-depth understanding of soil investigation through Standard Penetration Tests (SPT), concrete quality assessment through slump, compaction factor, and cube strength tests, and asphalt production through batching and bitumen mixing plant operations.

Exposure to safety systems, including Hazard Identification and Risk Assessment (HIRA), Third Party Inspection (TPI) of cranes, permit-to-work systems, and emergency response planning, strengthened practical awareness of construction safety and compliance. Quality assurance procedures across material handling, plant operations, and structural works enhanced understanding of construction standards and durability requirements under NHAI specifications.

Overall, the internship developed strong technical competence, field awareness, and professional discipline. It provided valuable insight into coordination among engineers, contractors, and workers, and significantly contributed to professional growth, preparing for future responsibilities in civil, highway, and structural engineering practice.